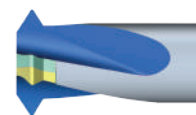
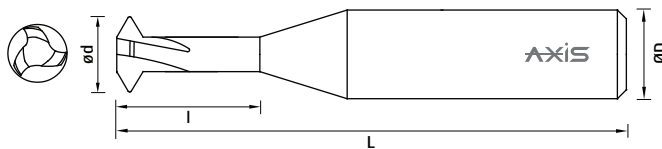
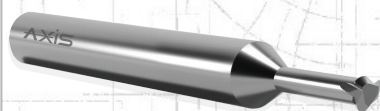


Solid Carbide Thread Mill



- Smooth cutting
- High efficient machining
- High strength edge shape design angle pitch
- Ultra precision thread whirling

Solid Carbide Thread Mill F5001



APPLICATION

A versatile tool designed for Threading on Steel Alloys, Stainless Steel, Titanium and Super Alloys in Medical, Digital Dentistry and Automotive segments

FEATURES & BENEFITS

Precisely controlled dimensions for accurate thread

Made from ultra fine grade carbide with a balanced mix of high toughness and excellent hardness

Cutting edge preparation by MMP Superfinishing process for superior edge strength, better chip evacuation, better coating adhesion and longer tool life

Metric Thread

Single Pitch

MC	MF	Pitch	ød	Neck Length l	L	ØD h6	Z
M0.6X0.150		0.150	0.43	1.70	38.00	3.00	3
M0.7X0.175		0.175	0.50	2.00	38.00	3.00	3
M0.8X0.200		0.200	0.57	2.30	38.00	3.00	3
M0.9X0.225		0.225	0.64	2.60	38.00	3.00	3
M1X0.25	—	0.250	0.71	2.90	38.00	3.00	3
—	M1X0.20	0.200	—	—	38.00	3.00	3
M1.2X0.25	M1.4X0.25 M1.6X0.25	0.250	0.91	3.40	38.00	3.00	3
M1.4X0.30	—	0.300	1.06	3.90	38.00	3.00	3
M1.6X0.35	—	0.350	1.20	4.50	38.00	3.00	3

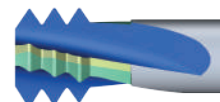
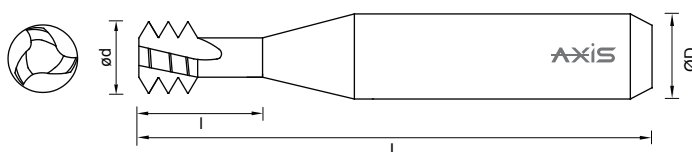
UN Thread

Single Pitch

UNC	UNF	TPI	ød	Neck Length l	L	ØD h6	Z
—	#0-80	80	1.15	4.30	38.00	3.00	3
#1-64	#2-64	64	1.45	5.10	38.00	3.00	3

World of mICRO Tools

Solid Carbide Thread Mill F5021



APPLICATION

A versatile tool designed for Threading on Steel Alloys, Stainless Steel, Titanium and Super Alloys in Medical, Digital Dentistry and Automotive segments

Metric Thread

3 Pitch

MC	MF	Pitch	Ød	Neck Length l	L	ØD h6	Z
M1.8X0.35	M2X0.35	0.35	1.40	5.00	38.00	3.00	3
M2X0.40	—	0.40	1.54	5.60	38.00	3.00	3
M2.5X0.45	—	0.45	1.98	6.90	38.00	3.00	3
M3X0.50	—	0.50	2.43	8.40	50.00	4.00	4
M3.5X0.60	—	0.60	2.81	9.90	50.00	4.00	4
M4X0.70	—	0.70	3.20	11.30	50.00	6.00	4
M5X0.80	—	0.80	4.08	14.00	50.00	6.00	4
M6X1.00	M8X1.00	1.00	4.90	16.80	50.00	6.00	4

FEATURES & BENEFITS

Precisely controlled dimensions for accurate thread

Designed for threading at faster feed

Made from ultra fine grade carbide with a balanced mix of high toughness and excellent hardness

Cutting edge preparation by MMP Superfinishing process for superior edge strength, better chip evacuation, better coating adhesion and longer tool life

UN Thread

3 Pitch

UNC	UNF	TPI	Ød	Neck Length l	L	ØD h6	Z
—	#1-72	72	1.45	5.10	38.00	3.00	3
#2-56	—	56	1.66	6.10	38.00	3.00	3
#4-40	—	40	2.11	7.90	50.00	4.00	3
—	#4-48	48	2.23	7.90	50.00	4.00	3
#5-40	#6-40	40	2.47	9.00	50.00	4.00	4
#6-32	—	32	2.59	10.20	50.00	4.00	4
#8-32	#10-32	32	3.25	11.90	50.00	6.00	4
—	#8-36	36	3.35	11.90	50.00	6.00	4
#10-24	5/16"-24	24	3.60	14.00	50.00	6.00	4
—	#10-32	32	3.91	14.00	50.00	6.00	4
1/4"-20	—	20	4.89	18.20	50.00	6.00	4
—	1/4"-28	28	4.95	18.20	50.00	6.00	4
5/16"-18	—	18	5.95	23.00	63.00	6.00	4
—	5/16"-24	24	5.95	23.00	63.00	6.00	4

World of mICRO Tools

• UNC : Unified National Coarse Thread

• UNF : Unified National Fine Thread

Cutting Parameters F5001 / F5021

Materials			Dia Range		
			0.30 - 1.00	1.01 - 3.00	3.01 - 6.00
P	Low/Unalloyed Steel <600 N/mm ²	Vc m/min	50-56	50-56	50-56
		Fz mm/tooth	0.004-0.01	0.01-0.05	0.02-0.10
	Low/Unalloyed Steel 600-1000 N/mm ²	Vc m/min	50-56	50-56	50-56
		Fz mm/tooth	0.004-0.01	0.01-0.05	0.02-0.10
	High Alloyed Steel 700-1500 N/mm ²	Vc m/min	50-56	50-56	50-56
		Fz mm/tooth	0.004-0.01	0.01-0.05	0.02-0.10
M	Ferritic/ Martensitic	Vc m/min	20-26	20-26	20-26
		Fz mm/tooth	0.004-0.010	0.01-0.03	0.02-0.05
	Austenitic	Vc m/min	20-26	20-26	20-26
		Fz mm/tooth	0.004-0.010	0.01-0.03	0.02-0.05
	Duplex/PH > 800 N/mm ²	Vc m/min	20-26	20-26	20-26
		Fz mm/tooth	0.004-0.010	0.01-0.03	0.02-0.05
K	GCI/NCI <180 BHN	Vc m/min	68-74	68-74	68-74
		Fz mm/tooth	0.004-0.010	0.01-0.05	0.04-0.10
	GCI/NCI 200-250 BHN	Vc m/min	50-56	50-56	50-56
		Fz mm/tooth	0.004-0.010	0.01-0.05	0.04-0.10
S	Ti-Based	Vc m/min	12 - 15	12 - 15	12 - 15
		Fz mm/tooth	0.004-0.010	0.01-0.03	0.02-0.05
	Fe, Ni, Co-Based	Vc m/min	10 - 15	10 - 15	10 - 15
		Fz mm/tooth	0.004-0.010	0.01-0.03	0.02-0.05

Ø 0.43 – 1.45 mm Art : F5001

Case Study Solid Carbide Thread Mill

Product Features

- Precisely controlled dimensions for accurate thread
- Made from ultra fine grade carbide with a balanced mix of high toughness and excellent hardness
- Cutting edge preparation by MMP Superfinishing process for superior edge strength, chip evacuation, better coating adhesion and longer tool life

Application

Material	SS304
Machine	VMC Makino
Component	Watch Case

Performance

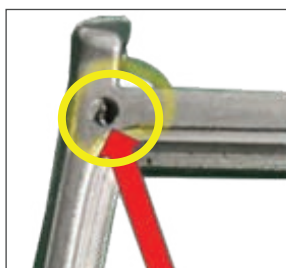
Tool Description : Ø0.91 x 3.40 x 39.00 x 3.00mm		
Tool Used	Competitor	AXIS
Thread Dimensions	M1.2x0.25P	M1.2x0.25P
Coating	Yes	Yes
Post Coat Treatment	N/A	Yes
Cutting Speed	25 m/min	25
RPM	8500	8500
Feed	250 mm/min	250
Burr	Visible	No burr better surface finish
Life (Components)	250	500

Accomplishment

- Increased tool life
250 to 500 components
- CPC reduced from 18.8 INR to 2.6 INR reduced by 7 times
- Ease of alignment between back cover and case components
- Burr free component and better finish

Wear Analysis

Competitor



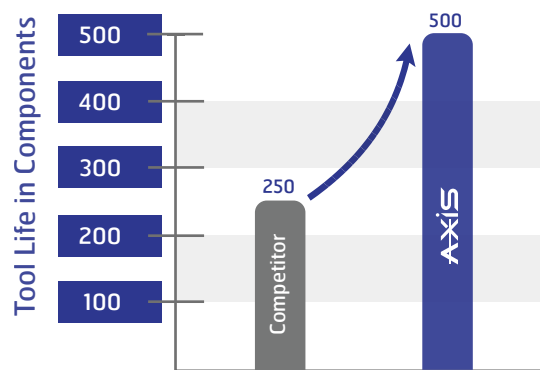
Burr on component

AXIS



Burr free component
and better finish

Results





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